



LAL
BAHADUR
SHASTRI

Group of Institutions

Lecture-6

“VB.Net Lab”

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Advanced Technology (Dot Net) Lab

1. Familiarization with IDE.
2. Programming Console applications using VB.NET covering all the aspects

of VB.NET fundamentals

3. Object oriented programming using VB.NET covering objects, Inheritance, Polymorphism,

Constructors, Static Classes, and Interfaces.

4. Programme to illustrate Exception Handling concepts.

5. Programme to illustrate use of Collections.

6. Programme to perform File I/O Operations.

7. Programming Windows applications using VB.NET covering all major controls and components,

Menus, MDI, Event Handling.

8. Creating windows installer.

9. Programme to interact with Database from a Windows Desktop Application.

10. Programming to build web applications using web controls, maintaining state.

Note: The Instructor may add/delete/modify/tune experiments, wherever he/she feels in a justified manner.

Introduction

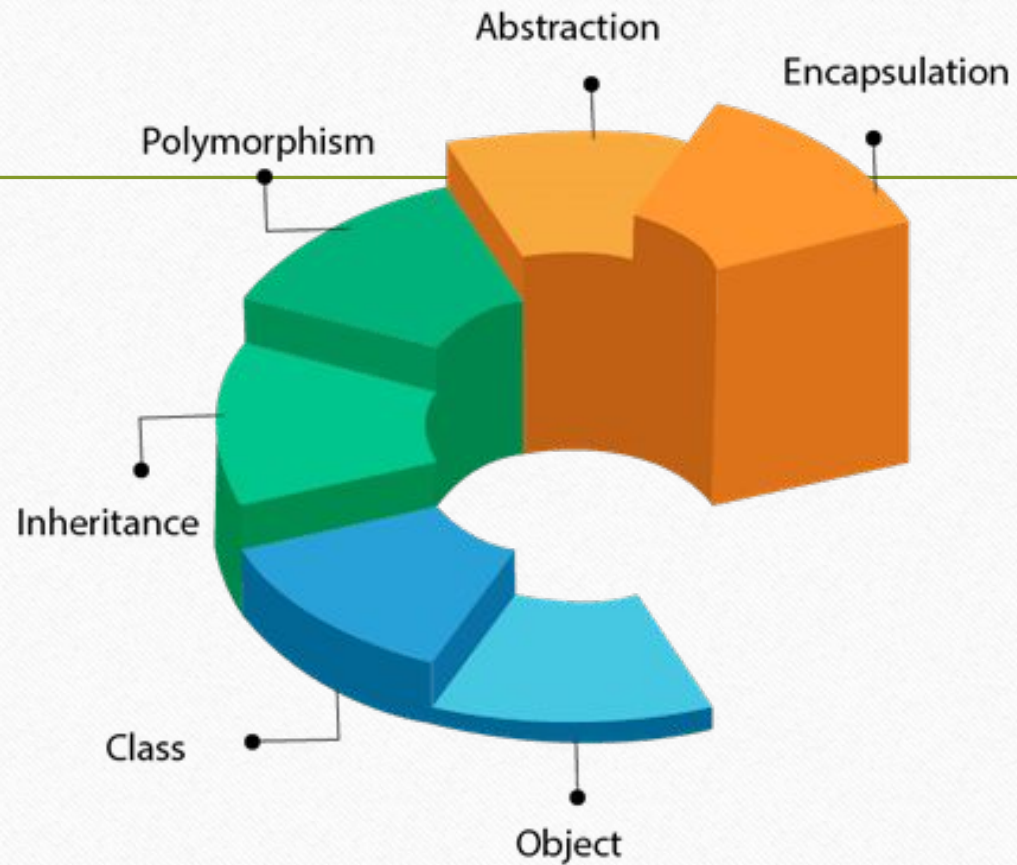
- Object-Oriented Programming is a paradigm that provides many concepts, such as **inheritance**, **data binding**, **polymorphism**, etc.
- **Simula** is considered the first object-oriented programming language. The programming paradigm where everything is represented as an object is known as a **truly object-oriented programming language**.
- **Smalltalk** is considered the first truly object-oriented programming language.

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- The popular object-oriented languages are Java, C#, PHP, Python, C++, etc.
 - The main aim of object-oriented programming is to implement real-world entities, for example, object, classes, abstraction, inheritance, polymorphism, etc.

OOPs (Object-Oriented Programming System)

- **Object** means a real-world entity such as a pen, chair, table, computer, watch, etc.
- **Object-Oriented Programming** is a methodology or paradigm to design a program using classes and objects. **It simplifies software development and maintenance** by providing some concepts:
 - Object
 - Class
 - Inheritance
 - Polymorphism
 - Abstraction
 - Encapsulation

OOPs (Object-Oriented Programming System)



Object

- Any entity that has state and behavior is known as an object. For example, a chair, pen, table, keyboard, bike, etc. It can be physical or logical.
- An Object can be defined as an instance of a class. An object contains an address and takes up some space in memory. Objects can communicate without knowing the details of each other's data or code. The only necessary thing is the type of message accepted and the type of response returned by the objects.
- **Example:** A dog is an object because it has states like color, name, breed, etc. as well as behaviors like wagging the tail, barking, eating, etc.

Class



- *Collection of objects* is called class. It is a logical entity.
- A class can also be defined as a blueprint from which you can create an individual object. Class doesn't consume any space.

• HumanBeing

• Male Female

Inheritance

- *When one object acquires all the properties and behaviors of a parent object, it is known as inheritance.*
- It provides code reusability. It is used to achieve runtime polymorphism.

Polymorphism

- If *one task is performed in different ways*, it is known as polymorphism.
- **For example: to convince the customer differently, to draw something, for example, shape, triangle, rectangle, etc.**
- In Java, we use method overloading and method overriding to achieve polymorphism.
- Another example can be to speak something; for example, a cat speaks meow, dog barks woof, etc.

Abstraction

- *Hiding internal details and showing functionality* is known as abstraction.
- For example phone call, we don't know the internal processing.
- In Java, we use abstract class and interface to achieve abstraction.

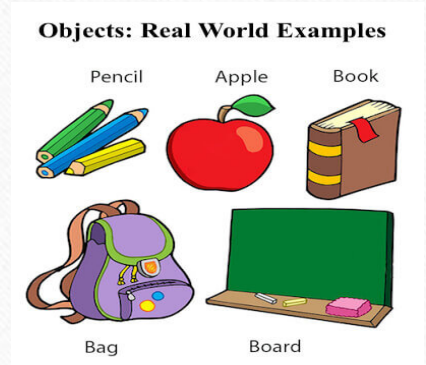
Encapsulation



Capsule

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- *Binding (or wrapping) code and data together into a single unit are known as encapsulation.* For example, a capsule, it is wrapped with different medicines.
 - A java class is the example of encapsulation. Java bean is the fully encapsulated class because all the data members are private here.

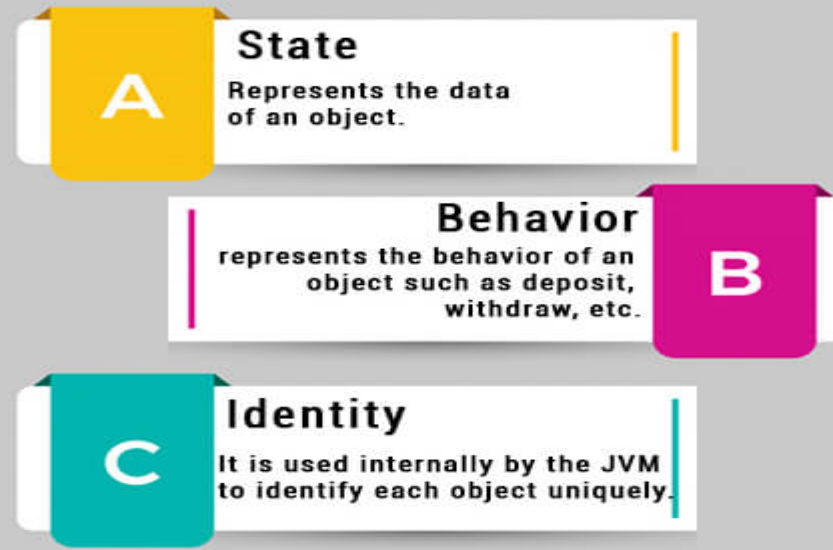
object



- An entity that has state and behavior is known as an object e.g., chair, bike, marker, pen, table, car, etc.
- It can be physical or logical (tangible and intangible).
- The example of an intangible object is the banking system.

- An object has three characteristics:
- **State:** represents the data (value) of an object.
- **Behavior:** represents the behavior (functionality) of an object such as deposit, withdraw, etc.
- **Identity:** An object identity is typically implemented via a unique ID. The value of the ID is not visible to the external user. However, it is used internally by the JVM to identify each object uniquely.

Characteristics of Object



For Example, Pen is an object. Its name is Reynolds; color is white, known as its state. It is used to write, so writing is its behavior.

Class

- A class is a template or blueprint from which objects are created. So, an object is the instance(result) of a class.

- A class is a group of objects which have common properties. It is a template or blueprint from which objects are created. It is a logical entity. It can't be physical.
- A class in Java can contain:
 - **Fields**
 - **Methods**
 - **Constructors**
 - **Blocks**
 - **Nested class and interface**

Syntax to declare a class:

```
class <class_name> {  
    field;  
    method;  
}
```


Instance variable in Java

- A variable which is created inside the class but outside the method is known as an instance variable.
- Instance variable doesn't get memory at compile time. It gets memory at runtime when an object or instance is created. That is why it is known as an instance variable.

Method in Java

- In Java, a method is like a function which is used to expose the behavior of an object.
- Code Reusability
- Code Optimization

new keyword

- The new keyword is used to allocate memory at runtime.
- All objects get memory in Heap memory area.

Thank You